

Relations & Equivalence

2.AB PreIB Maths – Exam C

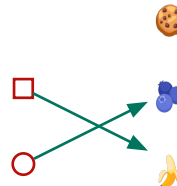
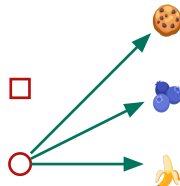
Unless specified otherwise, you are to **always** (at least briefly) explain your reasoning. Even in closed questions.

Product of Sets & Relations

Let $A = \{\text{🍌}, \text{🍌🍌}\}$ and $B = \{\text{🍌}, \text{🍇}, \text{🍪}\}$. Check all **correct** statements of the ones below. You **don't have to** explain yourself. [25 %]

- ☐ The set $\{(\text{🍪}, \text{🍌🍌}), (\text{🍪}, \text{🍇}), (\text{🍇}, \text{🍌})\}$ is a relation on B .
- ☐ The product $A \times B$ has the **same number of elements** as $B \times A$.
- ☐ The set $\{(\text{🍌}, \text{○}), (\text{○}, \text{🍪}), (\text{🍌}, \text{□})\}$ is a relation from B to A .
- ☐ The element $(\text{○}, \text{□})$ lies in $A \times A$.
- ☐ There are **more** relations on B than there are relations from A to B .

A relation F from A to B (A and B are the same as above) is called a **function** if every element $a \in A$ is **related** to **exactly one** element $b \in B$. Below, you see pictures of two relations from A to B . Are they functions? If not, **why**? [25 %]



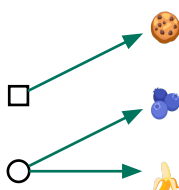
Bonus Problem

The **composition** of relations R from A to B and S from B to C is the relation written as $R \circ S$ from A to C that is defined by [10 %]

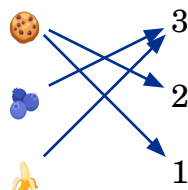
$$a(R \circ S)c \text{ in case that } aRb \text{ and } bSc$$

for any elements $a \in A, b \in B, c \in C$. Draw the picture for the relation $R \circ S$ if the pictures of R and S are as below.

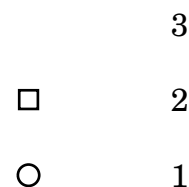
R



S



Draw $R \circ S$ here.



Equivalence

Recall that the relation of *equivalence* is given by three conditions:

[25 %]

- **reflexivity**: every element is equivalent to itself;
- **symmetry**: if a is equivalent to b , then b is equivalent to a ;
- **transitivity**: if a is eq. to b and b is eq. to c , then a is eq. to c .

Determine if the relation

$$R = \{(1, 1), (2, 2), (4, 4), (3, 4), (4, 2)\}$$

is an equivalence on the set $A = \{1, 2, 3, 4\}$. If not, add **as few pairs as possible** to make it into an equivalence. **Explain.**

Describe (as a set of pairs, by a picture, ...) an equivalence on the set $A = \{1, 2, 3, 4\}$

[25 %]

which has the following **classes of equivalence**:

- $\{1\}$, $\{2\}$ and $\{3, 4\}$
- $\{1, 2, 4\}$ and $\{3\}$.

You **don't need to** explain anything.

Bonus Problem

Prove that if $R \circ R \subseteq R$, then the relation R is **transitive**. See previous page for the definition of the composition of relations. The expression $R \circ R \subseteq R$ literally says that every pair in the composition of R with R also lies in R .

[10 %]